

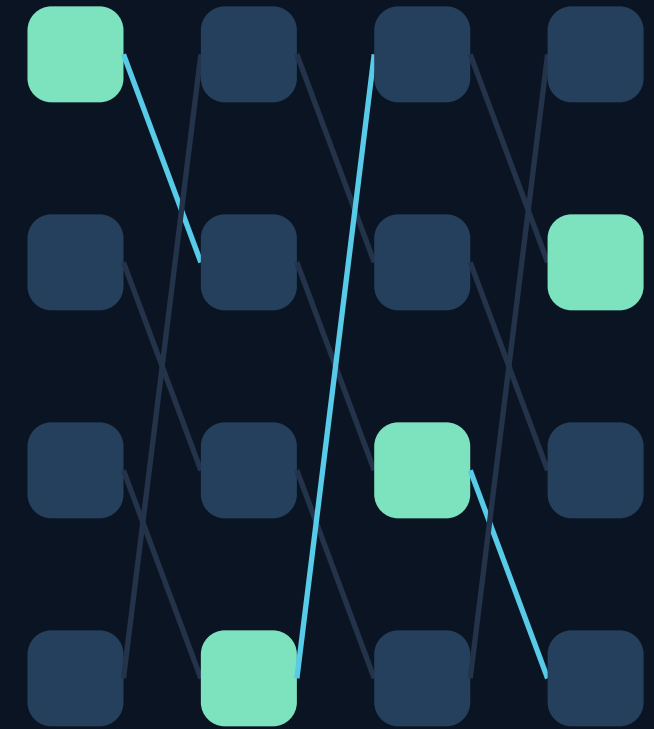
Sleeping Machines

AI that learns what to compute, when to compute it, and what to remember.

A new trainable architecture that already beats Transformer baselines at a fraction of the compute.

Tero Keski-Valkama | Sole founder

Raising €3M to scale research and engineering
Research-stage deep tech · 4 October 2026



Capacity beyond activity

Beats the saved Transformer baselines at a fraction of the compute.

Completed text8 language comparisons, same data and held-out test set (T256). Lower bits per character is better. Single seeds.

Comparison	Reference	Ours	Compute we used
vs Transformer-256×4, 4 passes (10M)	1.908	1.888	0.40× its training · 0.18× its inference
vs Transformer-256×2, 1 pass (10M)	2.427	1.955	0.96× its training · 0.16× its inference
vs LSTM-256 at matched inference (10M)	2.171	1.955	0.88× its inference (more training)
vs tuned LSTM-384, 6 passes (10M, first P0-6 arm)	1.840	1.888	1.39× its training: a loss; tuning both sides next
90M: LSTM-512 / Transformer (multi-pass)	1.661 / 1.604	1.857	≤1/16 of their training; matched runs queued
NeuroBench Mackey-Glass (20/30 repeats, sMAPE)	13.37 (LSTM)	14.37	57.6 KB model vs 490 KB

Matched-compute wins against both saved Transformers; a tuned small LSTM still beats us. Closing that is the next proof.

Native compute traced; references shape-estimated (saved E64 controls). Tuned dense baselines at our exact budgets are running for review. Scoreboard: README.md.

Model capacity and the cost of using it must scale differently.

The opportunity: preserve useful intelligence while reducing the work paid for each prediction.

MODEL CAPACITY

Know more

Larger models and richer memory can improve capability. Accessing that capacity still costs computation and data movement.

DEPLOYMENT ECONOMICS

Do less work

Serving cost, latency and memory determine whether capability can be deployed in constrained environments.

OUR APPROACH

Select + retain

Learn which local computations to activate; retain local state and train the alternatives that were not selected.

We are developing a model family and execution substrate around this separation.

One model family across today's architectural landscape.

Stateful temporal programs connect representation, memory, timing and learning.

Recurrent / SSM models

Persistent state and selective dynamics.
Add addressed temporal programs.

Attention / Transformers

Content-addressed aggregation.
Include rich local interaction.

Sparse experts / MoE

Capacity beyond selected expert work.
Add private facts and timing.

SLEEPING MACHINES / STATEFUL TEMPORAL PROGRAMS

One content / time / address / state interface; compatible choices from units to complete systems.
Integrated target: computational time, selective state, small messages, separate keys/values and unrealized credit.

Statistics / retrieval

Counts and direct historical evidence.
Combine with learned abstractions.

Adaptive memory / TTT

Learning in the deployed model.
Use local, causal update policies.

Event / temporal systems

Local arrivals, timers and causal order.
Add vector messages and route credit.

Containment needs actual operators and contracts. Learned parity, lower total cost and physical energy require evidence.

Choose the right computation in each region.

One family can mix selective temporal programs with rich interaction and causal memory.

01 / UNITS

Local programs

Choose state, flow, writes, small messages, keys and reception. Time can perform computation.



02 / LAYERS

Adaptive work

Select one or many messages. Add dense interaction where needed; preserve lean regions elsewhere.



03 / COMPOSITIONS

Shared evidence

Link learned vectors, counts and historical memory. Explicit paths let dense and sparse inputs inform one query.



04 / SYSTEMS

Same learner

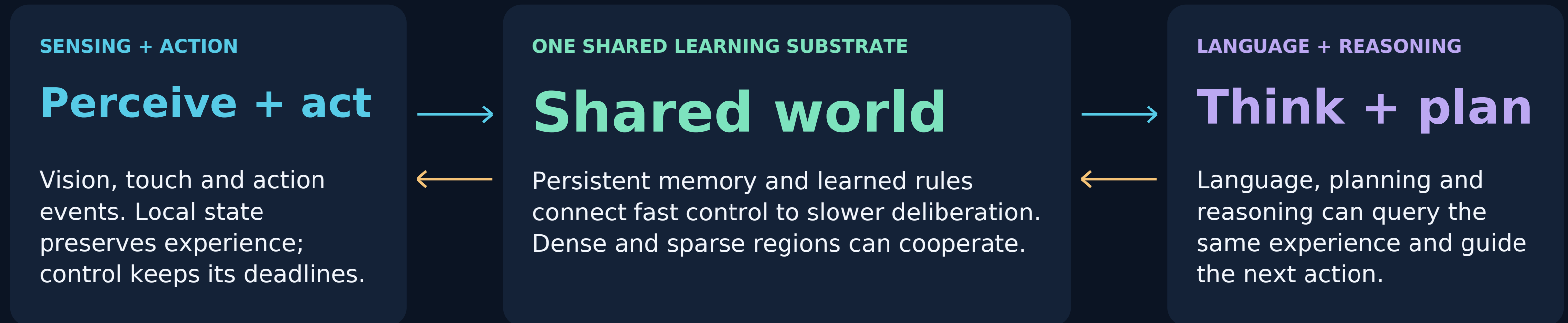
Support streamed or batched updates and native TTT. Broader structural design is a further learning ambition.

Counterfactual credit: learn from selected and unrealized routes

No required global periodic tick. Joint integration, measured TTT gains, morphing and hardware energy remain milestones.

A substrate for intelligence that learns across mind and body.

The ambition: connect sensing, action, language and reasoning in one trainable model, across their natural timescales.



Hypothesis: physical experience teaches reusable abstractions; reasoning improves action. Online learning keeps both adapting.

Upside: persistent generalist agents, from datacenter software to embodied intelligence on mobile robots.

First proof: held-out transfer in both directions, with retained skills and matched data/work. AGI is an ambition, not a demonstrated capability.

Messages race. Selected memories update. Alternatives learn.

A simple event loop connects representation, timing and sparse computation.

01 / ENCODE

Signal + delay

A learned function produces a message and its arrival time from input and local memory.



02 / SELECT

Temporal race

Arrival times select a receiver. Separate address keys and delivered values support sparse execution.



03 / REMEMBER

Local state

Update the selected memory and propagate a new message into deeper computation.



04 / LEARN

Alternatives

Use errors to improve message content and timing, including paths that were not selected.

Counterfactual credit: learn from selected and unrealized routes

Implemented language core: hard selection and persistent local state in a synchronous emulator.

The language experiments establish three useful facts.

10M-character text8 training budget; held-out prediction error. Lower bits per character is better.

IT CAN LEARN REAL TEXT

2.162 bits/char

Best model reaches the saved recurrent baseline's quality (2.171). It uses 2.89× estimated training work.

ALTERNATIVE CREDIT HELPS

−0.135 bits/char

In the same small model, learning from unchosen paths improves prediction error; estimated training work rises 0.30%.

CAPACITY AND SELECTED WORK

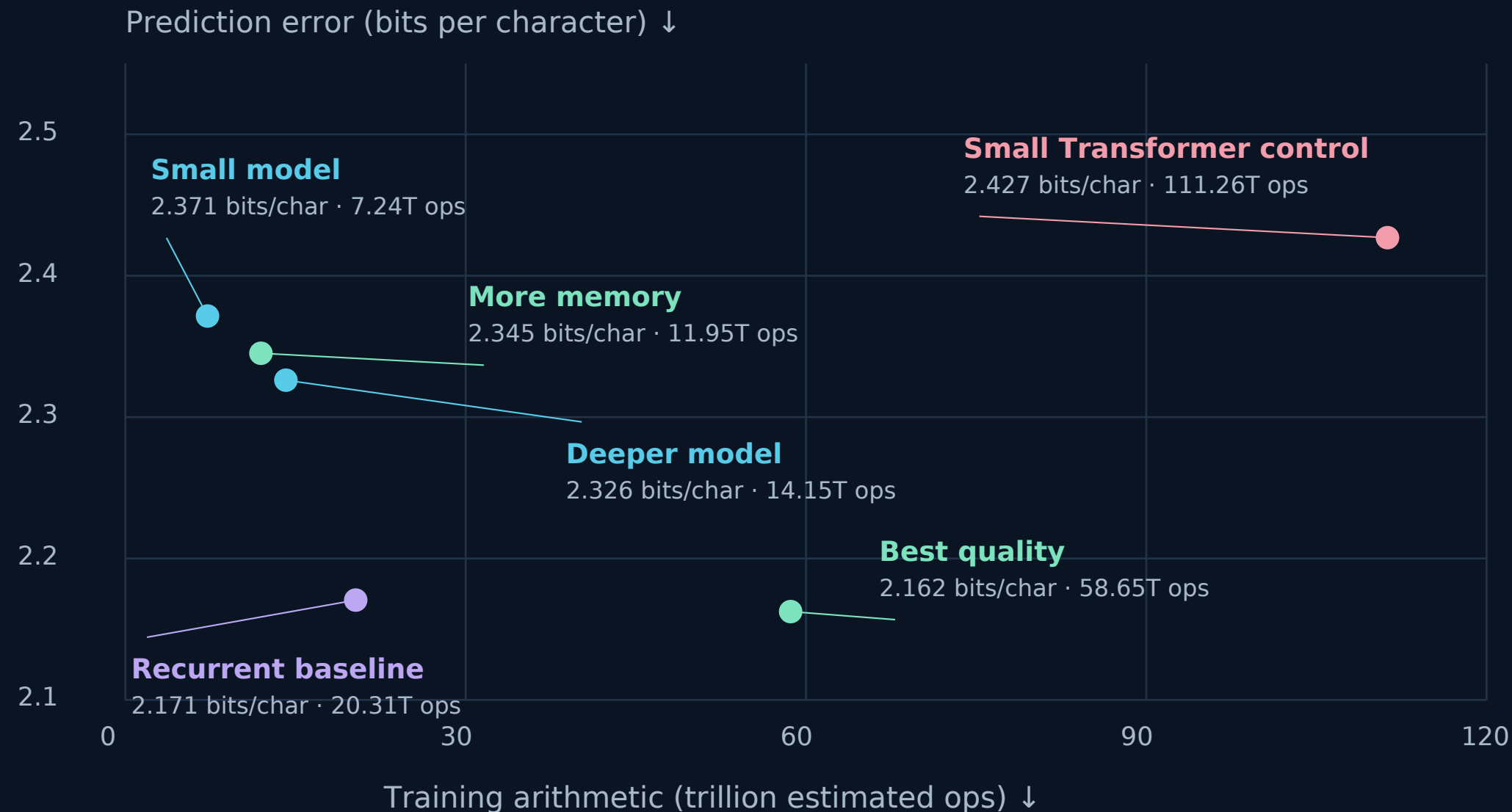
16 → 32 slots

More available memory improves prediction while retaining eight selected writes per input. Key scoring and learning costs also grow.

Exploratory single-seed results. The next proof is replicated quality plus a measured system-cost advantage.

A language-model foothold for a new architecture.

Held-out prediction error versus estimated training arithmetic; 10M characters, one pass.



QUALITY CHECK

2.162

Near recurrent baseline (2.171); 2.89× training work.

TRAINING ARITHMETIC

15.4× less

Small model versus saved Transformer control; better prediction error.

Saved one-pass controls; same test region and window length. Training order and work conventions differ. One seed. Arithmetic estimates are not service cost.

Start with low-batch language inference on existing hardware.

Product hypothesis for local and on-premise text applications; no customer interest reported yet.

WHO WOULD EVALUATE IT

Platform teams

Teams deploying small language models under latency, memory or cost constraints. Demand validation is a next step.

WHAT WE WOULD SUPPLY

Model + runtime

A trained model, sparse inference runtime and a reproducible quality-and-cost evaluation on the intended workload.

WHY THEY WOULD ADOPT

Verified savings

Adoption depends on equivalent useful quality, service reliability and a measured reduction in complete deployment cost.

Commercial gate: target $\geq 20\%$ full-system cost reduction at matched quality on one selected workload.

Software proves the advantage. Hardware multiplies it.

One research program, with staged routes to deployment.

01 / MODELS + RUNTIME

First product

Validate the language workload on existing hardware. License a model/runtime with auditable deployment economics.

02 / MEMORY-LOCAL ACCELERATORS

5.8× fewer bytes

Modeled per character: our 1.888-bpc model moves 5.8× fewer bytes than the Transformer it beats. 10× more capacity adds only 5% traffic. Data movement dominates chip energy.

03 / CLOCKLESS SILICON

Work $\propto$ information

No global clock: idle capacity draws only leakage, latency is average-case, and low average power permits denser 3D memory-on-logic packing. Also the missing model for the neuromorphic chip class.

The €3M round funds scale and a hardware cost model of the trained network; silicon follows measured proof.

A useful alternative could strengthen a hardware ecosystem.

Potential evaluation and licensing routes; no partner interest or relationship is claimed.

AMD

Model + platform

Its ~\$665M Silo AI acquisition and Liquid AI collaboration show the value of model/software capability on AMD hardware.

INTEL

Event execution

Hala Point and Loihi 2 provide a relevant research environment for expressive event models and continuous learning.

GOOGLE

Full-stack design

TPU model-hardware co-design creates a demanding evaluation venue for quality, memory and execution improvements.

Partner proposition: demonstrate a differentiated workload advantage that improves platform adoption.

Price the runtime against verified customer value.

Illustrative annual license economics; assumptions to test with an evaluation partner.

Per-customer assumption	Calculation	Annual amount
Eligible workload cost	Existing deployment cost we can actually influence	€1.00M
Gross savings	20% complete-cost reduction at matched quality	€200k
Runtime license	20% of verified gross savings	€40k

Customer retains €160k/year before migration and other costs. At the same assumed price, 25 deployments would generate €1M annual license revenue.

Validate pricing through matched quality, reliable operation and measured customer benefit.

AI research and software architecture in one founder.

Tero Keski-Valkama · Sole founder · Public career and invention record.

Tero Keski-Valkama

Sole founder

Experience spans applied ML, multimodal foundation models and software architecture. The round adds specialist capacity around the founder.

2012-2018

Cybercom · Software Architect

Historical role in public professional profile.

2018-2022

HERE · Lead ML & AI Engineer

Self-reported research and applied ML leadership.

2022-2025

Alloy.ai → Kaiko.ai

Staff engineering; multimodal foundation-model work.

2023 publication

EP4148389A2 · Named inventor

Maps from geospatial observations; applicant HERE.

Historical roles are self-reported; inventorship is documented. Planned hiring: learning research, runtime engineering, evaluation and measurement.

Build defensibility across the model and its execution.

The strongest potential asset is an integrated stack that is difficult to reproduce and deploy.

MODEL + LEARNING

Research know-how

Timing, state and alternative-credit methods, supported by retained ablations and numerical contracts.

RUNTIME + HARDWARE

Execution know-how

Sparse kernels, memory management, measurement and hardware mapping can reinforce the model advantage.

RIGHTS + INTEGRATION

Company assets

Establish contribution rights and licensing, then develop tooling and partner-specific deployment expertise.

Private development supports controlled disclosure. IP ownership and commercial integrations remain diligence items.

Three gates determine whether this becomes a business.

Each gate has a falsifiable result and a consequence for further spending.

01 / QUALITY AT SCALE

Replicate

At least three seeds, fixed causal tests and tuned controls. Compare quality and complete work; keep failures beside the best result.

02 / COMPLETE ECONOMICS

Measure

Pass trained sparse parity. Measure total serving cost, tail latency, memory and energy at useful quality on one fixed workload.

03 / ADOPTION

Evaluate

Identify a use case and an evaluator willing to test it. Confirm useful quality, integration effort and willingness to pay.

If a gate fails, diagnose or narrow the workload before committing more compute or new silicon.

The next 18 months produce evidence investors can underwrite.

Proposed milestones measured from funding; spending advances with completed gates.

MONTHS 0-3

Team + proofs

Hire the initial team.
Establish company/IP rights. Complete trained sparse execution checks and larger-data, multi-seed evaluation.

MONTHS 3-6

One workload

Run tuned controls and measure total cost, latency, memory and energy. Select a workload only if quality and economics hold.

MONTHS 6-12

External trial

Seek independent reproduction and prospective customer evaluations. Test pricing, integration effort and willingness to pay.

MONTHS 12-18

Product + scale

Turn a validated workload into a deployable runtime. Expand models and assess hardware mapping for the next financing stage.

Next-round objective: replicated quality, a measured cost advantage and a credible evaluation pathway.

€3M funds an 18-month research and engineering program.

A proposed six-person average team, staged compute and hardware feasibility.



- Team: research + systems €1.35M | 45.0%
- Compute + independent replication €0.90M | 30.0%
- Hardware feasibility + measurement €0.25M | 8.3%
- IP, legal, operations €0.20M | 6.7%
- Contingency reserve €0.30M | 10.0%

€2.70M operating plan + €0.30M reserve · Planned operating burn averages €150k/month.
Budget assumptions: six average FTE including founder at €150k/year fully loaded; compute and hiring quotes remain to be obtained.

Raise €3M to establish a commercially valuable platform.

Proposed priced round; terms remain subject to company formation and investor diligence.

CAPITAL TO RAISE

€3M

18 months of research, systems engineering and staged validation.

PROPOSED PRE-MONEY

€50M

A premium research-stage proposal for the model/runtime/substrate opportunity.

INITIAL OWNERSHIP

5.66%

€53M post-money, before option-pool changes, preferences and later dilution.

The proposal prices a model/learning family, sparse runtime and future compute substrate that already beats Transformer baselines at a fraction of the compute. Step-up path: a confirmed 90M matched-compute win and a NeuroBench first place support the €100M scenario.

Valuation sensitivities and conditional exit scenarios are in the appendix.

Build the model. Prove the economics. Open the hardware path.

€3M proposed raise · €50M proposed pre-money · Private investor discussion.

€3 million

A new architecture is becoming trainable.
This round tests whether its advantage scales.

- 01** Replicated language quality and learning mechanisms.
- 02** A deployable runtime with measured system economics.
- 03** External evaluation and a credible path to hardware.

Tero Keski-Valkama · Private investor diligence available by arrangement